

Skill 7:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>
#include <sys/stat.h>

int main(int argc, char *argv[])
{
    pid_t pid;
    int status;

    printf("=====\n");
    printf("PROCESS SYNCHRONIZATION USING waitpid()\n");
    printf("=====\n");

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());

        sleep(2);

        printf("Child Process Completed\n");
        exit(0);
    }
    else
    {

```

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printf("\n=====\\n");
printf("PATH VARIABLE AND COMMAND RESOLUTION\\n");
printf("=====\\n");

char *path = getenv("PATH");

if (path == NULL)
{
    printf("PATH variable not found.\\n");
    return 1;
}

printf("PATH Variable:\\n%s\\n\\n", path);

char *path_copy = strdup(path);
char *directory = strtok(path_copy, ":");

char command[100];

if (argc > 1)
{
    strcpy(command, argv[1]);
}
else
{
    strcpy(command, "ls");
}

printf("Searching for command: %s\\n", command);

int found = 0;

while (directory != NULL)
{
    char full_path[500];

    snprintf(full_path, sizeof(full_path),

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    snprintf(full_path, sizeof(full_path),
             "%s/%s", directory, command);

    struct stat file_info;

    if (stat(full_path, &file_info) == 0)
    {
        printf("Executable found: %s\n", full_path);

        if (access(full_path, X_OK) == 0)
        {
            printf("Execute permission: YES\n");
            found = 1;
            break;
        }
        else
        {
            printf("Execute permission: NO\n");
        }
    }

    directory = strtok(NULL, ":");
}

if (!found)
{
    printf("Command '%s' not found in PATH.\n", command);
}

free(path_copy);

printf("\nCommand resolution test completed.\n");

return 0;
}
```

```

bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p ~/process_waitpid
cd ~/process_waitpid
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ pwd
/home/bhanu/process_waitpid
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ nano waitpid_path.c
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ gcc waitpid_path.c -o waitpid_path
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ ls
waitpid_path  waitpid_path.c
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ ./waitpid_path
=====
PROCESS SYNCHRONIZATION USING waitpid()
=====
Parent Process Started
Parent PID: 916
Child Process Started
Child PID: 917
Child Process Completed
Child exited normally with status: 0
Parent Process Continues After Child Completion

=====
PATH VARIABLE AND COMMAND RESOLUTION
=====
PATH Variable:
/home/bhanu/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0.x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/PuTTY:/mnt/c/Users/Admin/AppData/Local/Agv/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Admin/.local/bin:/mnt/c/Users/Admin/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Local/Python/bin:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Users/Admin/AppData/Local/Programs/Antigravity IDE/bin:/snap/bin

Searching for command: ls
Executable found: /usr/bin/ls
Execute permission: YES

Command resolution test completed.
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ ./waitpid_path cat
=====
PROCESS SYNCHRONIZATION USING waitpid()
=====
Parent Process Started
Parent PID: 918
Child Process Started
Child PID: 919
Child Process Completed
Child exited normally with status: 0
Parent Process Continues After Child Completion

=====
PATH VARIABLE AND COMMAND RESOLUTION
=====
PATH Variable:
/home/bhanu/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0.x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/PuTTY:/mnt/c/Users/Admin/AppData/Local/Agv/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Admin/.local/bin:/mnt/c/Users/Admin/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Local/Python/bin:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Users/Admin/AppData/Local/Programs/Antigravity IDE/bin:/snap/bin

Searching for command: cat
Executable found: /usr/bin/cat
Execute permission: YES

Command resolution test completed.
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ ./waitpid_path gcc
=====
PROCESS SYNCHRONIZATION USING waitpid()
=====
Parent Process Started
Parent PID: 920
Child Process Started
Child PID: 921
Child Process Completed
Child exited normally with status: 0
Parent Process Continues After Child Completion

=====
PATH VARIABLE AND COMMAND RESOLUTION
=====
PATH Variable:
/home/bhanu/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0.x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/PuTTY:/mnt/c/Users/Admin/AppData/Local/Agv/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Admin/.local/bin:/mnt/c/Users/Admin/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Local/Python/bin:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Users/Admin/AppData/Local/Programs/Antigravity IDE/bin:/snap/bin

Searching for command: gcc
Executable found: /usr/bin/gcc
Execute permission: YES

Command resolution test completed.
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ ./waitpid_path abcdxyz
=====
PROCESS SYNCHRONIZATION USING waitpid()
=====
Parent Process Started
Parent PID: 922
Child Process Started
Child PID: 923
Child Process Completed
Child exited normally with status: 0
Parent Process Continues After Child Completion

=====
PATH VARIABLE AND COMMAND RESOLUTION
=====
PATH Variable:
/home/bhanu/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0.x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/PuTTY:/mnt/c/Users/Admin/AppData/Local/Agv/bin:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Admin/.local/bin:/mnt/c/Users/Admin/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Admin/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Admin/AppData/Local/Python/bin:/mnt/c/Users/Admin/AppData/Roaming/npm:/mnt/c/Users/Admin/AppData/Local/Programs/Antigravity IDE/bin:/snap/bin

Searching for command: abcdxyz
Command 'abcdxyz' not found in PATH.

Command resolution test completed.
bhanu@DESKTOP-DE32HQ8:~/process_waitpid$ |

```